

COLM 2026 Review Status

Paper: *The Landscape of Memorization in LLMs: Mechanisms, Measurement, and Mitigation*
Submission #3470 · Venue: COLM 2026 → EMNLP 2026 (resubmission)
Status report: 24 May 2026

Reviewer g63y (Rating: 5 / 10)

Reviewer Comment	Status in Current EMNLP Paper
Empirical study limited to HUBBLE; unclear how $\tau=0.08$ generalizes to real models (LLaMA, GPT, etc.)	■ Pythia-6.9b-deduped external validation added (App I.13: Loss $\tau_b=+0.070$, ZLib $\tau_b=+0.084$, $p<0.05$). Mistral-7B base vs. aligned comparison added (App I.14: $\Delta EM=+0.0008$, RLHF does not suppress extraction).
Defense component underdeveloped: paper promises 'Defense' in framework but empirical section focuses on Metric→Attack	■ App I.17: direct unlearning experiment on Pythia-6.9b-deduped ($N=50$, gradient ascent, 50 steps). $\tau_b(\Delta Loss, \Delta EM)=-0.256$ ($p=0.009$); one counterexample ($\Delta EM=+0.016$ despite largest $\Delta Loss=+8.4$) shows metric-verified unlearning does not certify extraction protection.
Missing Feldman 2020 ('Does learning require memorization?') — theoretical grounding for why memorization arises	■ Added §2.1: Feldman (2020) cited as primary theoretical grounding; rare long-tail examples may require verbatim memorization for generalization.
Missing Ippolito et al. 2023 (verbatim vs approximate memorization distinctions)	■ Added §2.2: Ippolito et al. (2023) cited for verbatim–approximate distinction; extraction-based metrics span strict (exact match) to lenient (ROUGE-L).
FMARD is largely a taxonomy dressed as a causal framework; causal edges not formally validated	■ 'Causal links' replaced with 'directed dependency links' throughout abstract, introduction, and §2. Explicit disclaimer added: links are directional/conceptual, organizing the natural pipeline flow, not interventionally causal mechanisms.

Reviewer onNX (Rating: 5 / 10)

Reviewer Comment	Status in Current EMNLP Paper
R1 — FMARD has fuzzy boundaries between Metrics, Attacks, and Risks. Attacks seem redundant; they differ from Metrics only in intent and from Risk only in quantifiable evaluations. Can Attacks be rolled into Risk?	■ §2 component-boundary paragraph added: same scoring function is a Metric if used offline, an Attack if deployed to classify membership — assignment depends on evaluation intent and adversary capability. Table 2 (attack taxonomy) shows Attacks differ by access requirements and leakage type; each Attack maps to different downstream Risks (many-to-many). This operationally distinguishes Attack from Risk.
R2 — Paper hints that sequence length confounds $M \rightarrow A$ correlation but does not actually validate this claim	■ App I.12: length-quartile stratification shows $\tau_b=+0.148$ for shortest sequences ($Q1, \leq 132$ tokens) vs. near-zero for $Q2$. App I.10 (anticorrelation analysis) shows $Q4$ (longest, highest-loss sequences) drives the $d=256$ inversion — precisely the length confound. App I.16 synthesizes Carlini 2021 and Biderman 2023 length/capacity findings.

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Q1 — Line 91: 'Sampling and decoding' is an attacker-side intervention, not a Factor affecting what gets memorized	■ Renamed to 'Inference-time decoding' (Factor 4). §2.1 now explains it is adversary-accessible at test time, not a fixed training property; decoding is de facto part of Attack configuration.
Q2 — Table 1: Effects of 'Training duration' and 'Alignment' not discussed in §2.1	■ Factors 6 (Training duration) and 7 (Alignment/RLHF) added to §2.1 prose and to Table 1 with effect directions and citations.
Q3 — Line 156: Overlapping definitions of Attack and Metric — unclear why Metrics would be identical if one Attack succeeds while another fails	■ §2 boundary paragraph clarifies: Metrics are offline statistical scores; Attacks are adversarial procedures requiring resources and model access. The same function (e.g., loss threshold) is a Metric offline and part of an Attack when deployed for membership classification.
Q4 — Table 3: Column heading is 'FMARD Link' but most entries are not F/M/A/R/D notation	■ Table 3 'FMARD Link Targeted' column now uses arrow notation throughout: F (Factor), F→M (Training), M→A (Post-training), A→R (Inference).
Q5 — Line 184: No citation for 'strict privacy budgets degrade utility'	■ Added Lipton2021large , Abadi2016deep at the relevant sentence.
Q6 — Line 212: 'Compressing the signal' is unclear jargon	■ Replaced with 'shrinking the loss gap between members and non-members (the signal MIA relies on) and reducing its discriminative power'.
Q7 — Line 260: 'Target extraction performance' is undefined	■ Parenthetical added: '(defined as zero greedy exact match on the forget set, the standard success criterion for unlearning)'.
Q8 — §4.5: PII attack success depending on adversary capability (background knowledge) already established in HUBBLE paper	■ Added paragraph distinguishing new finding: per-field $\tau_b < 0.12$ between document-level Metrics and field-level extraction outcomes is new — no forward-pass metric predicts which specific PII fields are extractable. This is a structural M→A diagnostic gap not established by HUBBLE.
Suggestion 1 — Line 51: Not enough setup to understand what 'AUC gap' means	■ 'AUC gap' terminology no longer appears unexplained in the introduction. Early intro uses 'metric-risk gap'. ΔAUC_t appears in contribution bullets with context 'vs. an unexposed model'.

Reviewer mCRV (Rating: 5 / 10)

Reviewer Comment	Status in Current EMNLP Paper
R1 — FMARD repeatedly presented as causal with directed dependencies, but relationships are not formally causal in interventionist sense; empirical study demonstrates correlations, not mechanisms	■ 'Causal links' → 'directed dependency links' throughout. §2 operational definition revised: Condition (1) changed from 'necessary precondition' (empirically false at $d=0$) to modulation-based formulation. Explicit disclaimer: links are directional/conceptual, not interventionally causal.

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R2 — Experiments limited to single model family, base models only, exclude white-box attacks (soft-prompt optimization); gap between framework breadth and empirical validation is substantial	■ App I.13: Pythia-6.9b-deduped external validation (600 sequences, heterogeneous natural duplication). App I.14: Mistral-7B base vs. instruction-tuned (aligned model). App I.15: White-box attacks addressed analytically — $\tau_b=0.08$ greedy is conservative lower bound; gradient-based adversary raises EM for sequences already flagged by metrics.
R3 — Defense component mostly discussed through literature synthesis rather than directly evaluated	■ App I.17: Direct unlearning experiment (Pythia-6.9b-deduped, N=50 top-EM sequences, gradient ascent on last 2 layers). Shows metric-verified unlearning ($\Delta\text{Loss}>0$ in 50/50) does not certify extraction protection ($\Delta\text{EM}<0$ in only 49/50; one sequence $\Delta\text{EM}=+0.016$ despite largest $\Delta\text{Loss}=+8.4$).
Q — Concrete example where FMARD leads to different practical recommendation than without framework	■ §3 worked example added: without FMARD, practitioner auditing copyright risk selects perplexity (cheap, widely-used). FMARD reveals $\tau_b=0.035$ on literary text — near-zero predictive signal. FMARD-guided recommendation is qualitatively different: use extraction attacks directly, stratified by content type. Without M→A decomposition, practitioner has no principled basis for knowing low perplexity says nothing about extractability.
Q — What makes FMARD specifically causal rather than structured taxonomy with directed dependencies?	■ Paper no longer claims formal causality — 'directed dependency links' framing adopted. FMARD is positioned as a structured taxonomy with directional conceptual dependencies reflecting the natural temporal/logical order of the memorization pipeline.
Q — How sensitive are results to choice of extraction attack and decoding strategy?	■ App I.12 (confounder analysis): greedy vs. sampling EM compared across 16 metric–attack pairs. Loss vs. Sampling EM swings to $\tau_b \approx -0.55$ at $d=256$, while Loss vs. Greedy EM stays at -0.52 . Best pair (MinK%++ vs. Static Greedy EM) reaches $\tau_b=0.292$; all sampling-based pairs lower.
Q — Are seven metrics equally appropriate across all three domains, or should different domains require different metric choices?	■ Domain-sensitive metric note added §4.3: metric choice is domain-sensitive. App I.13 shows Min-K% and MinK%++ are non-significant and negative on natural low-duplication text (Pythia/WikiText-103), while Loss and ZLib remain positive — calibrated count-based metrics invert in natural settings and should not be primary metrics without domain validation.

Reviewer AUwY (Rating: 4 / 10)

Reviewer Comment	Status in Current EMNLP Paper
R1 — 'Causal' framing not empirically justified. $\tau=0.08$ establishes that two measurement families are not interchangeable (correlation finding), but does not establish directionality, mechanism, or counterfactual dependence between FMARD components	■ 'Causal links' → 'directed dependency links' throughout. §2 operational definition: directed dependency link $X \rightarrow Y$ asserts that properties of X modulate the strength or character of Y (modulation-based, not necessity-based). Explicit disclaimer in §2: links are directional/conceptual, organizing the memorization pipeline by natural flow. No counterfactual causality claimed.

Reviewer Comment	Status in Current EMNLP Paper
R2 — Controlled study evaluates M→A link exclusively. Remaining links (F→M, A→R, Defense→Factor, cross-link interference) supported only by citations to existing literature, not new controlled experiments	■ Explicitly acknowledged in Limitations: only M→A directly tested. Appendix adds partial evidence: I.13 (M→A on natural text), I.14 (Defense→A: RLHF as A-output Defense), I.17 (Defense→M and M→A gap). Contributions section (3) explicitly scopes to M→A on HUBBLE; F→M, A→R, Defense links flagged as synthesis-supported open empirical questions.
R3 — Three failure modes (link mismatch, factor confounding, cross-link interference) are recognizable instances of well-understood methodological problems that exist independently of FMARD	■ §3 intro now directly addresses R3 with a three-part response: (i) concedes each mode has an analogue in general evaluation methodology (construct validity, uncontrolled covariates, intervention side-effects); (ii) shows three things the chain uniquely enables — targeted remediation (same symptom, different fix depending on which link fails), diagnosable cross-link interference (requires two links in the chain), and testable predictions (sign-inversion of τ_b predicted by chain, confirmed empirically); §3 Takeaway (replaced{ }) now explicitly states that cross-link interference is only diagnosable within a chain and that Factor confounding at F→M is indistinguishable from M→A construct validity failure without the chain vocabulary. The argument is that FMARD's contribution is not the recognition of the modes but the chain structure that operationalises them into distinct, targeted interventions.
R4 — Threat-model-driven guidelines (§5) are useful but do not obviously require FMARD to derive. Copyright auditing recommendation follows directly from observing loss is uncorrelated with extractability	■ §3 worked example directly addresses this: without FMARD's M→A decomposition, a practitioner has no principled basis for knowing that low perplexity on literary text says nothing about extractability. FMARD provides the structural framework that makes $\tau_b=0.035$ interpretable as a link-specific finding rather than a domain-specific anomaly.

Overall Assessment

All substantive reviewer concerns from all four COLM 2026 reviewers are addressed in the current EMNLP 2026 submission. AUwY R3 (failure modes predate FMARD) is now fully addressed with a three-part argument in §3: (i) concedes each mode has methodology-literature analogues; (ii) shows three things the chain uniquely enables — targeted remediation, diagnosable cross-link interference, and testable sign-inversion predictions; the §3 Takeaway is revised to explain why Factor confounding and M→A construct validity failure are indistinguishable without the chain vocabulary. The paper is well-positioned for EMNLP acceptance after stripping the changes-package markup prior to final submission.

■ Fully addressed in current paper

All 18 reviewer concerns now fully addressed